Complex factors

1 Complete the following factorings:

a $9ix - 54 = 9i\left(\square\right)$

b $3ix - 18 = 3i\left(\square\right)$

c $8ix + 32 = 8i\left(\square\right)$

d $6ix + 30 = 6i\left(\square\right)$

2 Factor the following expressions. Leave your answers in terms of i.

a $x^2 + 25$

b $x^2 + 36$

c $x^2 + 100$

d $x^2 + 64$

e $x^2 + 4$

f $x^2 + 81$

g $x^2 + 121$

h $x^2 + 169$

3 Factor the following expressions. Leave your answers in terms of i.

a $64x^2 + 81$

b $3x^2 + 108$

c $25x^2 + 4$

d $16x^2 + 49$

e $36x^2 + 144$

f $100x^2 + 25$

g $81x^2 + 49$

h $16x^2 + 64$

4 Factor the following expressions. Leave your answers in terms of i

a $x^2 + 12ix - 36$

b $x^2 + 8ix - 16$

c $x^2 + 10ix - 25$

d $x^2 + 4ix - 4$

e $x^2 + 2ix - 1$

f $x^2 + 6ix - 9$

g $x^2 + 20ix - 100$

h $x^2 + 14ix - 49$

5 Factor the following expressions. Leave your answers in terms of i

a $x^2 + 10ix - 16$

b $x^2 + 8ix - 15$

c $x^2 + 7ix - 12$

d $x^2 + 11ix - 30$

e $x^2 + 12ix - 32$

f $x^2 + 14ix - 40$

g $x^2 + 13ix - 40$

h $x^2 + 18ix - 77$

6 Factor the following expressions. Leave your answers in terms of i

a $64x^2 + 80ix - 25$

b $49x^2 + 56ix - 16$

c $4x^2 + 36ix - 81$

d $16x^2 + 40ix - 25$

e $36x^2 + 36ix - 9$

f $25x^2 + 60ix - 36$

g $9x^2 + 24ix - 16$

h $100x^2 + 140ix - 49$

7 Factor the following expressions. Leave you[redacted]wers in terms of i

a $15x^2 + 59ix - 56$

b $2x^2 + 17ix - 21$

c $6x^2 + 11ix + 4$

d $8x^2 + 42ix - 54$

e $16x^2 - 22ix + 3$

f $12x^2 + 11ix - 5$

g $2x^2 - 15ix - 27$

h $35x^2 + 57ix - 18$